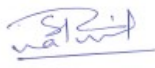


	FIR VEHICLE TESTING		11/08/2025
FIR No :FIR/Testing-Outdoor-UA/520100/2025/00720			
Part No/ Group :52010160R			
Subject :Report on Fuel Economy evaluation of Signa 3522.T CNG BSVI-II (with AC ON & OFF) vehicle in comparison with Signa 3523.T BSVI-II AC OFF base vehicle.			
Test Vehicle	Project Name :Signa 3522.T CNG BSVI-II Model Name : Signa 3522.T CNG BSVI-II (		Enclosures : Annexure 1-5
	Chasis Number :PC053604042RL1028 Engine Number :HVX106918		
FIR Status :	ESO FIR		

Role	Name	Signature
Prepared By	Palash Das	 11-Aug-2025
Checked By	Swapnil Jadhav	 12-Aug-2025
Approved By	Ajaykumar Jadhav	 12-Aug-2025

Cc: Head CVBU Engg, Chief Engineer, Head COC, Project Team [Design]; Project Team [Testing]

PA20 F222 (R4) March, 2021

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Documents : Please click here

[FIR_Testing_Outdoor_UA_52..](#)

 FIR/Testing-Outdoor-UA/520100/2025/00720	FIR VEHICLE TESTING	Date: 10/07/2025		
Part No / Group: 52010160R	Specification no :- " 259052070002 "		Action by	Failure class

Subject: -	Report on Fuel Economy evaluation of Signa 3522.T CNG BSVI-II (with AC ON & OFF) vehicle in comparison with Signa 3523.T BSVI-II AC OFF base vehicle.		
1.0 <u>Conclusion:</u>			
1.1 <u>Fuel economy in laden (rated load) at 40 kmph with AC ON & OFF condition:</u>			
Fuel economy of Signa 3522.T CNG BSVI-II vehicle found to be 4.03 kmpkg (with AC ON), which is better by 11.79% & 4.21 kmpkg (with AC OFF) which is better by 15.64% wrt Signa 3523.T BSVI-II base Vehicle.		--	--
1.2 <u>Fuel economy in Unladen at 40 kmph with AC ON & OFF condition:</u>			
Fuel economy of Signa 3522.T CNG BSVI-II vehicle found to be 6.79 kmpkg (with AC ON), which is better by 4.48% & 7.25 kmpkg (with AC OFF) which is better by 9.30% wrt Signa 3523.T BSVI-II base Vehicle.		--	--
1.3 <u>Overall fluid economy status at 40 kmph (80% Laden + 20% unladen condition):</u>			
Fluid economy of Signa 3522.T CNG BSVI-II vehicle found to be 4.38 kmpkg (with AC ON), which is better by 10.85% & 4.60 kmpkg (with AC OFF) which is better by 14.84% wrt Signa 3523.T BSVI-II base Vehicle.		--	--
1.4 <u>Fuel economy in laden (rated load) at 55 kmph with AC ON & OFF condition:</u>			
Fuel economy of Signa 3522.T CNG BSVI-II vehicle found to be 3.90 kmpkg (with AC ON), which is better by 13.11% & 4.00 kmpkg (with AC OFF) which is better by 17.82% wrt Signa 3523.T BSVI-II base Vehicle.		--	--
1.5 <u>Fuel economy in Unladen at 55 kmph with AC ON & OFF condition:</u>			
Fuel economy of Signa 3522.T CNG BSVI-II vehicle found to be 6.01 kmpkg (with AC ON), which is better by 4.10% & 6.65 kmpkg (with AC OFF) which is better by 15.90% wrt Signa 3523.T BSVI-II base Vehicle.		--	--
1.6 <u>Overall fluid economy status at 55 kmph (80% Laden + 20% unladen condition):</u>			
Fluid economy of Signa 3522.T CNG BSVI-II vehicle found to be 4.20 kmpkg (with AC ON), which is better by 11.85% & 4.34 kmpkg (with AC OFF) which is better by 17.57% wrt Signa 3523.T BSVI-II base Vehicle.		--	--
2.0 <u>Result:</u> Vide Observations			
FIR Status		Page 1 of 5	

 FIR/Testing-Outdoor-UA/520100/2025/00720	FIR VEHICLE TESTING		Date: 10/07/2025	
Part No / Group: 52010160R	Specification no :- " 259052070002 "		Action by	Failure class

FAILURES CLASS: A: Failure that happen without warning. Detrimental to vehicle / personnel, road users and safety. Can cause death / hospitalization. Noncompliance of regulatory requirement. B: Vehicle gets crippled with sufficient warning. Minor injury to occupant & road users (-- First aid only). Premature failures / repeated failures. Leads to consequential catastrophic damages. Need immediate attention. C: Customer irritants. Capable of damaging aesthetics. Failure can be attended at next servicing Schedule. D: Customer perception issues and minor irritants.

3.0 Test Vehicle: 1. Signa 3522.T CNG BSVI-II (ERC-1309) Chassis No. – PC053604042RL1028 Engine No. – HVX106918 Base Vehicle: 2. Signa 3523.T Ash BSVI-II (ERC-1309) Chassis No. – PC53562032RJ00873 Engine No. – 22L63919785 Input Doc Ref.: - Testing-Outdoor-UA/01948 DID No: – NA. DDP No: – NA.	Enclosures: Annexure 1: Panel chart Annexure 2: PAT Sheet Annexure 3: PAT Summary Annexure 4: Test request Annexure 5: Test plan
Work Order No. :- P.23.EA.504 Make :- TATA Motors Limited	T I No. UA001757

Earlier Ref: - NA

4.0 Project Status:- DR3

5.0 Introduction: As a part of CNG 5.3L program, Signa 3522.T CNG BSVI-II vehicle has been offered for AC ON & OFF Fuel Economy evaluation with Signa 3523.T BSVI-II AC OFF base Vehicle. This report provides a comprehensive overview of the trial details and its status.
--

6.0 Testing:

Following test was conducted during evaluation :

Sr. no.	Test	Reference standard	Instruments used
1	Fuel economy evaluation	TST/VT/WG/158	Cummins Data Logger

FIR Status	Page 2 of 5
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 FIR/Testing-Outdoor-UA/520100/2025/00720		FIR VEHICLE TESTING		Date: 10/07/2025		
Part No / Group: 52010160R		Specification no :- " 259052070002 "			Action by	Failure class

7.0 Vehicle details:

Parameter		Test vehicle
Vehicle model		Signa 3522.T CNG BSVI-II (ERC-1309)
Engine Model		TATA 5.3 SGI Bharat Stage VI OBD-II
Calibration ID		53SGITCBS6
CAL Used in Vehicle		53SGITCBS6
Max Power		220 HP @2300 RPM
Max Torque		850 Nm @ 1100 ~1600 rpm
GVW (kg)		35000
Gear Box		TATA G-950 DD
Gear Ratios.		1-9.36, 2- 5.20, 3- 3.05, 4- 1.84, 5- 1.33, 6- 1.00
Axle Ratio		5.85
Tyres	FA1	Size: : 295/90R 20, Make: M/s Apollo Endu, Type : Phase-II
	RA1	Size: : 295/90R 20, Make: M/s Apollo Endu, Type : Phase-II
	RA2	Size: : 295/90R 20, Make: M/s Apollo Endu, Type : Phase-II
Fan Details (Type/Make/Blades)		Type: E-Viscous fan and Size:24"

Parameter	Test vehicle
Vehicle model	Signa 3522.T CNG BSVI-II (ERC-1309)
Base Cal Improvement (Cal Maturity %)	90% Cal Maturity
Low Viscous Engine Oil (10W30)	--
FE Switch	Yes(E/B/P)
Engine Features (EOL : Enable/Disable)	Disable
ATS Optimization	--
APM	--
Cruise Control	No
Gear based VAM	No
Intelligent mode shifter	No
Moving Mass Estimation	No
Driveline Optimization	No
GSA (Gear Shift Advisor)	Yes(Not Tune)
MSA (Mode Shift Advisor)	--
Low CRR Tyres	Yes
Low Viscous rear axle oil (80W90)	Yes
Smart Alternator (LIN)	No
THU/Unitised/Low friction bearing	No

CALIBRATION ID: 53SGITCBS6

FIR Status

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	FIR VEHICLE TESTING	Date: 10/07/2025
FIR/Testing-Outdoor-UA/520100/2025/00720		
Part No / Group: 52010160R	Specification no :- " 259052070002 "	Action by Failure class

8.0 Test result:

8.1 Cruising speed at 40 km/hr with AC ON & OFF Condition:

Sr. No.	Parameters	Base Vehicle	Test Vehicle
1	Vehicle Identification No. (ERC No./EI No./Reg. No.)	Signa 3523.T 5.6L BSVI-II ERC-1309	Signa 3522.T.CNG BSVI-II EL00313
2	Vehicle GVW/GCW	35000	35000
3	Unladen weight	10500	10500
4	Engine Details (Make/cc/Norms/no. of Cyl.)	Cummins ISB6 5.6L/Ashwamedh 220 HP @ 2300 rpm	TATA 5.3 SGI Bharat Stage VI OBD-II 220 HP @ 2300 rpm
5	Engine Max Power	925 Nm @ 1000-1300 rpm	850 Nm @ 1100-1600 rpm
6	Engine Calibration No.	XWTA13007	535GTCB56
7	Engine Calibration Maturity Level	As per production released	90%
8	Regeneration Strategy (Time ass/Soot base)	Time Based (100 Hrs)	90%
9	Gear Box (make/drive/FGR)	G950 65 DD, FGR- 9.36	G950 65 DD, FGR- 9.36
10	Gear Ratios	1-9.36, 2- 5.20, 3- 3.05, 4- 1.84, 5- 1.33, 6- 1.00	1-9.36, 2- 5.20, 3- 3.05, 4- 1.84, 5- 1.33, 6- 1.00
11	BAR	5.12	5.85
12	Type Details	Bridgestone 295/90R20 low CRR	Apollo Endu NR9 295/90R20
13	LBSIC	High & Low Engine Speed Breakpoint Vehicle Mass Threshold	1900 rpm, 2000 rpm 35000
14	VAM	Variable Acceleration 1, 2 Variable Acceleration Speed 1, 2	10.2, 5 10, 40
15	GDP	GDP Max. Veh. Speed (Heavy & Light Engine Load)	50, 50
16	RSD	Maximum Accelerator Vehicle Speed	80
17	Torque	Torque Limit Medium Adjustment Factor	130%
18	Factor	Torque Limit Maximum Adjustment Factor	120%
19	PTP	Powertrain Protection (Enable / Disable)	Disable
20	Torque Limit	Disable	Disable
21	USFE / Torque Table Revision (1/2/3/m)	USFE-1	USFE-1
22	Fan Details (type/make/blades)	24" E-Viscous fan	24" E-Viscous fan
23	Base Cal Improvement (Cal Maturity %)	100%	100%
24	Low Viscous Engine Oil (TMGO-10W30)	Yes (15W40CK4+)	Yes (15W40)
25	FE Switch	USFE	Yes (3 Mode : E/B/P)
26	Engine Features (EOL : Enable/Disable)	Enable	No
27	ATS Optimization	Yes	No
28	APM	Yes	No
29	Gear based VAM	No	No
30	Intelligent mode shifter	No	No
31	Moving Mass Estimation	No	No
32	Driveline Optimization	No	No
33	GSA (Gear Shift Advisor)	Yes	Yes (Not Tuned)
34	MSA (Mode Shift Advisor)	No	No
35	Low CRR Tyres	Yes	Yes
36	Low Viscous rear axle oil (80W90)	Yes	Yes
37	Smart Alternator (LIN)	Yes	Yes
38	THU/Unlubed/Low friction bearing	No	No
39	Route	No	No
40	FE Duty Cycle (Laden/Unladen)	Laden Trial @ 40 kmph	Unladen Trial @ 40 kmph
41	Trial Date	30.05.25	31.05.25
42	Trip Distance	242.00	242.00
43	Total CNG quantity filled	66.25	63.53
44	Total Diesel Quantity	66.25	63.53
45	Total Linea Quantity	66.25	63.53
46	Equivalent Linea Quantity (100% Consumption)	66.25	63.53
47	Total Equivalent Fuel Consumed	66.25	63.53
48	Fuel Economy	3.65	3.84
49	Equivalent Fuel Economy	3.56	3.71
50	Status of Equivalent Fuel economy	Base	Base
51	Cluster Total Odo. Reading (at End of trial)	254	254
52	Cluster Trial AFE	3.90	4.00
53	Avg. Equivalent Fuel Economy	3.64	3.61
54	Status of Fuel Economy	Base	Base
55	Status of AC ON wrt AC OFF	Base	Base
56	Overall Avg. Fuel Economy (kmpl/kmplkg) : L + UL	4.00	3.96
57	Overall status of Avg. Fuel Economy (kmpl/kmplkg) : L + UL	80% + 20%	80% + 20%

8.2 Cruising speed at 55 km/hr with AC ON & OFF Condition:

Sr. No.	Parameters	Base Vehicle	Test Vehicle
1	Vehicle Identification No. (ERC No./EI No./Reg. No.)	Signa 3523.T 5.6L BSVI-II ERC-1309	Signa 3522.T.CNG BSVI-II EL00313
2	Vehicle GVW/GCW	35000	35000
3	Unladen weight	10500	10500
4	Engine Details (Make/cc/Norms/no. of Cyl.)	Cummins ISB6 5.6L/Ashwamedh 220 HP @ 2300 rpm	TATA 5.3 SGI Bharat Stage VI OBD-II 220 HP @ 2300 rpm
5	Engine Max Power	925 Nm @ 1000-1300 rpm	850 Nm @ 1100-1600 rpm
6	Engine Calibration No.	XWTA13007	535GTCB56
7	Engine Calibration Maturity Level	As per production released	90%
8	Regeneration Strategy (Time ass/Soot base)	Time Based (100 Hrs)	90%
9	Gear Box (make/drive/FGR)	G950 65 DD, FGR- 9.36	G950 65 DD, FGR- 9.36
10	Gear Ratios	1-9.36, 2- 5.20, 3- 3.05, 4- 1.84, 5- 1.33, 6- 1.00	1-9.36, 2- 5.20, 3- 3.05, 4- 1.84, 5- 1.33, 6- 1.00
11	BAR	5.12	5.85
12	Type Details	Bridgestone 295/90R20 low CRR	Apollo Endu NR9 295/90R20
13	LBSIC	High & Low Engine Speed Breakpoint Vehicle Mass Threshold	1900 rpm, 2000 rpm 35000
14	VAM	Variable Acceleration 1, 2 Variable Acceleration Speed 1, 2	10.2, 5 10, 40
15	GDP	GDP Max. Veh. Speed (Heavy & Light Engine Load)	50, 50
16	RSD	Maximum Accelerator Vehicle Speed	80
17	Torque	Torque Limit Medium Adjustment Factor	130%
18	Factor	Torque Limit Maximum Adjustment Factor	120%
19	PTP	Powertrain Protection (Enable / Disable)	Disable
20	Torque Limit	Disable	Disable
21	USFE / Torque Table Revision (1/2/3/m)	USFE-1	USFE-1
22	Fan Details (type/make/blades)	24" E-Viscous fan	24" E-Viscous fan
23	Base Cal Improvement (Cal Maturity %)	100%	100%
24	Low Viscous Engine Oil (TMGO-10W30)	Yes (15W40CK4+)	Yes (15W40)
25	FE Switch	USFE	Yes (3 Mode : E/B/P)
26	Engine Features (EOL : Enable/Disable)	Enable	No
27	ATS Optimization	Yes	No
28	APM	Yes	No
29	Gear based VAM	No	No
30	Intelligent mode shifter	No	No
31	Moving Mass Estimation	No	No
32	Driveline Optimization	No	No
33	GSA (Gear Shift Advisor)	Yes	Yes (Not Tuned)
34	MSA (Mode Shift Advisor)	No	No
35	Low CRR Tyres	Yes	Yes
36	Low Viscous rear axle oil (80W90)	Yes	Yes
37	Smart Alternator (LIN)	Yes	Yes
38	THU/Unlubed/Low friction bearing	No	No
39	Route	No	No
40	FE Duty Cycle (Laden/Unladen)	Laden Trial @ 55 Kmph	Unladen Trial @ 55 Kmph
41	Trial Date	28.05.25	29.05.25
42	Trip Distance	242.00	242.00
43	Total CNG quantity filled	70.46	70.00
44	Total Diesel Quantity	70.46	70.00
45	Total Linea Quantity	70.46	70.00
46	Equivalent Linea Quantity (100% Consumption)	70.46	70.00
47	Total Equivalent Fuel Consumed	70.46	70.00
48	Fuel Economy	3.43	3.46
49	Equivalent Fuel Economy	3.25	3.27
50	Status of Equivalent Fuel economy	Base	Base
51	Cluster Total Odo. Reading (at End of trial)	254	254
52	Cluster Trial AFE	3.70	3.80
53	Avg. Equivalent Fuel Economy	3.39	3.45
54	Status of Fuel Economy	Base	Base
55	Status of AC ON wrt AC OFF	Base	Base
56	Overall Avg. Fuel Economy (kmpl/kmplkg) : L + UL	3.69	3.75
57	Overall status of Avg. Fuel Economy (kmpl/kmplkg) : L + UL	80% + 20%	80% + 20%

FIR Status

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Format No. PA20 F206 (R4) Feb, 2021

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FIR/Testing-Outdoor-UA/520100/2025/00720

FIR VEHICLE TESTING

Date: 10/07/2025

Part No / Group: 52010160R

Specification no :- " 259052070002 "

Action
byFailure
class

9.0 Torque table:- Test Vehicle :

3522.T 5.7L BSVI-II (CNG) _Heavy Mode											
Heavy Mode		RAR	5.85								
G950	Gears	700	1000	1200	1400	1600	1800	2000	2200	2400	2750
9.36	1	295	800	850	850	850	810	740	685	635	0
5.2	2	295	800	850	850	850	810	740	685	635	0
3.05	3	295	800	830	830	830	810	740	685	635	0
1.84	4	295	800	830	830	830	810	740	685	635	0
1.33	5	295	800	830	830	830	810	740	685	635	0
1	6	295	800	830	830	830	810	740	685	635	0
3522.T 5.7L BSVI-II (CNG) _Medium Mode											
G950 DD	Gears	700	1000	1200	1400	1600	1800	2000	2200	2400	2750
9.36	1	295	800	850	850	850	810	740	685	635	0
5.2	2	295	800	850	850	850	810	740	685	635	0
3.05	3	295	800	850	850	850	810	740	685	635	0
1.84	4	295	685	685	685	685	660	615	575	530	0
1.33	5	295	685	685	685	685	660	615	575	530	0
1	6	295	685	685	685	685	660	615	575	530	0
3522.T 5.7L BSVI-II (CNG) _Light Mode											
G950 DD	Gears	700	1000	1200	1400	1600	1800	2000	2200	2400	2750
9.36	1	295	800	850	850	850	810	740	685	635	0
5.2	2	295	800	850	850	850	810	740	685	635	0
3.05	3	295	685	685	700	700	660	615	575	530	0
1.84	4	295	575	575	575	575	570	560	545	505	0
1.33	5	295	575	575	575	575	570	560	545	505	0
1	6	295	575	575	575	575	570	560	545	505	0

Base Vehicle

➤ Balance mode (TF – 130%) – Test Vehicle

	600	800	1000	1200	1400	1600	1800	2000	2200	2400	2600	2800
1	500	500	850	753	400	400	400	400	400	400	80	0
1.33	500	500	850	800	753	543	400	400	400	400	80	0
1.84	500	500	850	800	753	543	400	400	400	400	80	0
3.05	500	500	850	800	800	753	543	400	400	400	80	0
5.2	450	500	850	850	850	850	772	694	616	412	80	0
9.36	450	500	850	850	850	850	772	694	616	412	80	0

➤ Eco mode (TF – 120%) – Test Vehicle

	600	800	1000	1200	1400	1600	1800	2000	2200	2400	2600	2800
1	500	500	850	753	400	400	400	400	400	400	80	0
1.33	500	500	850	800	753	543	400	400	400	400	80	0
1.84	500	500	850	800	753	543	400	400	400	400	80	0
3.05	500	500	850	800	800	753	543	400	400	400	80	0
5.2	450	500	850	850	850	850	772	694	616	412	80	0
9.36	450	500	850	850	850	850	772	694	616	412	80	0

10.1 Further Testing: As per further development plan.

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 TATA MOTORS LTD. ERC,PUNE		PAT ATTRIBUTES SHEET - Fuel Economy										VE CVBU/FE/01						
		Depart ment Name: Vehicle Engineering Project Name: Signa 3522.T CNG BSV-II Max Power- 220 hp @ 2300 rpm Max Torque - 850 Nm 1100-1600 rpm Driveline : G950 DD 5.85 RAR 295/90 R20 Tyres 3 Mode FE Switch EOL Parameters : RSG: 80 kmph Base Vehicle- Signa 3523.T BSV-II										PAT Sheet No.: VE CVBU/ HCV/ Signa 3522.T CNG BSV-II / FE/01 Date: 14/07/2025						
Sr.No.	Aggregate and test parameter						Unit	Test Type	Classification @DR4	Ref.Standard / Work Guideline	Simulation	Base Vehicle Signa 3523.T BSV-II	Target (AC ON)	Test Vehicle	Status (RYG)	Remarks	CoC Remark	
	Level 2	Level 3	Level 4	Level 5	Level 6	Level 7								Signa 3522.T CNG BSV-II (G950 DD 5.85 RAR 295/90 R20)				
A	Fuel Economy Evaluation						Heading											
2.0	Fuel Economy At Rated Load						Heading											
2.7	Highway AC ON / OFF						Heading											
2.7.1	Route : Jamshedpur-Ranchi							Standard	Specification	TSTVTW/G/158		Iteration 1 With AC ON	Iteration 2 With AC OFF	Iteration 1 With AC ON	Iteration 2 With AC OFF			
2.7.9	Average Fluid Economy @40 kmph						Kmpkg/Kmpl		Target			3.6	3.64	± 3.60	4.03	4.21	OK	Signa 3522.T CNG BSV-II was tested 828 with Signa 3523.T BSV-II Base Vehicle on JBR-Ranchi route.
2.7.10	Average Fluid Economy @55 kmph						Kmpkg/Kmpl		Data Generation			3.45	3.39		3.90	4.00	--	
4.0	Fuel Economy At Unladen						Heading											
4.7	Highway AC ON / OFF						Heading											
4.7.1	Route : Jamshedpur-Ranchi							Standard	Specification	TSTVTW/G/158								
4.7.9	Average Fluid Economy @40 kmph						Kmpkg/Kmpl		Target			6.50	6.64	± 6.50	6.79	7.25	OK	Target has been taken from Base vehicle tested data.
4.7.10	Average Fluid Economy @55 kmph						Kmpkg/Kmpl		Data Generation			5.78	5.74		6.01	6.65	--	
5.0	Overall Fuel Economy (80%Laden + 20%Unladen)						Heading											
5.1	Highway AC ON / OFF						Heading											
5.1.1	Route : Jamshedpur-Ranchi							Standard	Specification	TSTVTW/G/158								
5.1.9	Average Overall Fluid Economy @40 kmph						Kmpkg/Kmpl		Target			3.69	4.00	± 3.69	4.38	4.60	OK	
5.1.10	Average Overall Fluid Economy @55 kmph						Kmpkg/Kmpl		Data Generation			3.75	3.69		4.20	4.34	--	
Sign Off																		
Responsibility												Index for Status						
Name												Off target, no resolution						
Signature												Marginally off target, resolution planned/low risk						
PAT Engineer Palash Das/ Swapnil Jadhav												Met target, no concern						
PAT Lead / Project Lead Ajaykumar Jadhav / Sanjeev kumar																		

Annexure-1

Level 1 Attribute:					PANEL Chart					VE CVBU/FE/01												
PAT- Fuel Economy					Specification: Max Power- 220 hp @ 2300 rpm Max Torque - 850 Nm 1100-1600 rpm Driveline : G950 DD 5.85 RAR 295/90 R20 Tyres 3 Mode FE Switch EOL Parameters : RSG: 80 kmph					Panel Chart No: VE CVBU/ HCV/ Signa 3522.T CNG BSVI-II / FE/01												
Programme : Signa 3522.T CNG BSVI-II Application			Gateway : DR3		Owner: Ajaykumar Jadhav				Project Name: Signa 3522.T CNG BSVI-II		Date: 11/07/2025											
Level 1 Summary:					Statement Of Intent (SOI) or DNA Description:																	
<table><tr><td rowspan="2">PALS Position</td><td colspan="2">Status</td></tr><tr><td>Current/ Gateway</td><td>Job#1 Predicted</td></tr><tr><td>Level 1:</td><td>L</td><td>L</td><td colspan="3"></td></tr></table>					PALS Position	Status		Current/ Gateway	Job#1 Predicted	Level 1:	L	L				Fuel Economy: Haulage Application						
PALS Position	Status																					
	Current/ Gateway	Job#1 Predicted																				
Level 1:	L	L																				

Deriv.	Attribute Target Summary:										Remarks	
	Level 2	Level 3	Level 4	Level 5	Level 6	PALS	Metric	Target (AC ON)	Current/Gateway Signa 3522.T CNG BSVI-II	Job#1 Predicted	Signa 3523.T BSVI-II Vehicle	
	Specification								Max Power-220 hp @ 2300 rpm Max Torque - 850 Nm @ 1100-1600 rpm Driveline : G950 DD 5.85 RAR 3 Mode FE Switch		Max Power-220 hp @ 2300 rpm Max Torque - 925 Nm @ 1300-1600 rpm Driveline : G950 DD 5.12 RAR	1. Signa 3522.T CNG BSVI-II was tested B2B with Signa 3523.T BSVI-II Base Vehicle on JSR-Ranchi route. 2. Target has been taken from Positive at-par TML base BS-VI RDE Non AC vehicle tested in back-to-back condition.
	Vehicle GVW (Payload)						kg		35000		35000	
	Power (Max) @ rpm						HP		220 hp @ 2300 rpm		220 hp @ 2300 rpm	
	Power to Weight ratio						HP/Ton		6.29		6.29	
	Torque (Max) @ rpm						Nm		850 Nm 1100-1600 rpm		925 Nm @ 1300-1600 rpm,	
	Torque to Weight ratio						Nm/Ton		24.29		26.43	
	Tyres size						--		Apollo 295/90 R20		Bridgstone 295/90 R20	
	Gear Box Ratios						--		G950 DD, FGR -9.36		G950 DD, FGR -9.36	
	Final Drive Ratio						--		5.85		5.12	
									Iteration-1 With AC ON	Iteration-2 With AC OFF	With AC ON	With AC OFF
	Fuel Economy at Rated Load					L			L			
	Highway AC ON / OFF						--					
	Route : Jamshedpur-Ranchi						--					
	Avg. Fuel Economy @40 kmph						kmpl	≥ 3.60	4.03	4.21	3.60	3.64
	Avg. Fuel Economy @55 kmph						--		3.90	4.00	3.45	3.39
	Fuel Economy At Unladen					L			L			
	Highway AC ON / OFF						--					
	Route : Jamshedpur-Ranchi						--					
	Avg. Fuel Economy @40 kmph						kmpl	≥ 6.50	6.79	7.25	6.50	6.64
	Avg. Fuel Economy @55 kmph						--		6.01	6.65	5.78	5.74
	Overall Fuel Economy (80% Laden + 20% Unladen)					L			L			
	Highway AC ON / OFF						--					
	Route : Jamshedpur-Ranchi						--					
	Overall Avg. Fuel Economy @40 kmph						kmpl	≥ 3.69	4.38	4.60	3.69	4.00
	Overall Avg. Fuel Economy @55 kmph						--		4.20	4.34	3.75	3.69

Executive Summary:	1. Signa 3522.T CNG BSVI-II was tested B2B with Signa 3523.T BSVI-II Base Vehicle on JSR-Ranchi route. 2. Target has been taken from Positive at-par TML base BS-VI RDE Non AC vehicle tested in back-to-back condition.										
	Tested by:					Verified and Approved by:					
 Palash Das / Swapnil Jadhav PAT ENGINEER					 Ajaykumar Jadhav / Sanjeev Kumar PAT LEAD / PROJECT LEAD						

OUR CULTURE CREDO

AT TATA MOTORS

We are connecting aspirations by being bold in thought and action, owning every opportunity and challenge, Solving together as one team and engaging all our stakeholders with empathy.

We are **MORE WHEN ONE!**

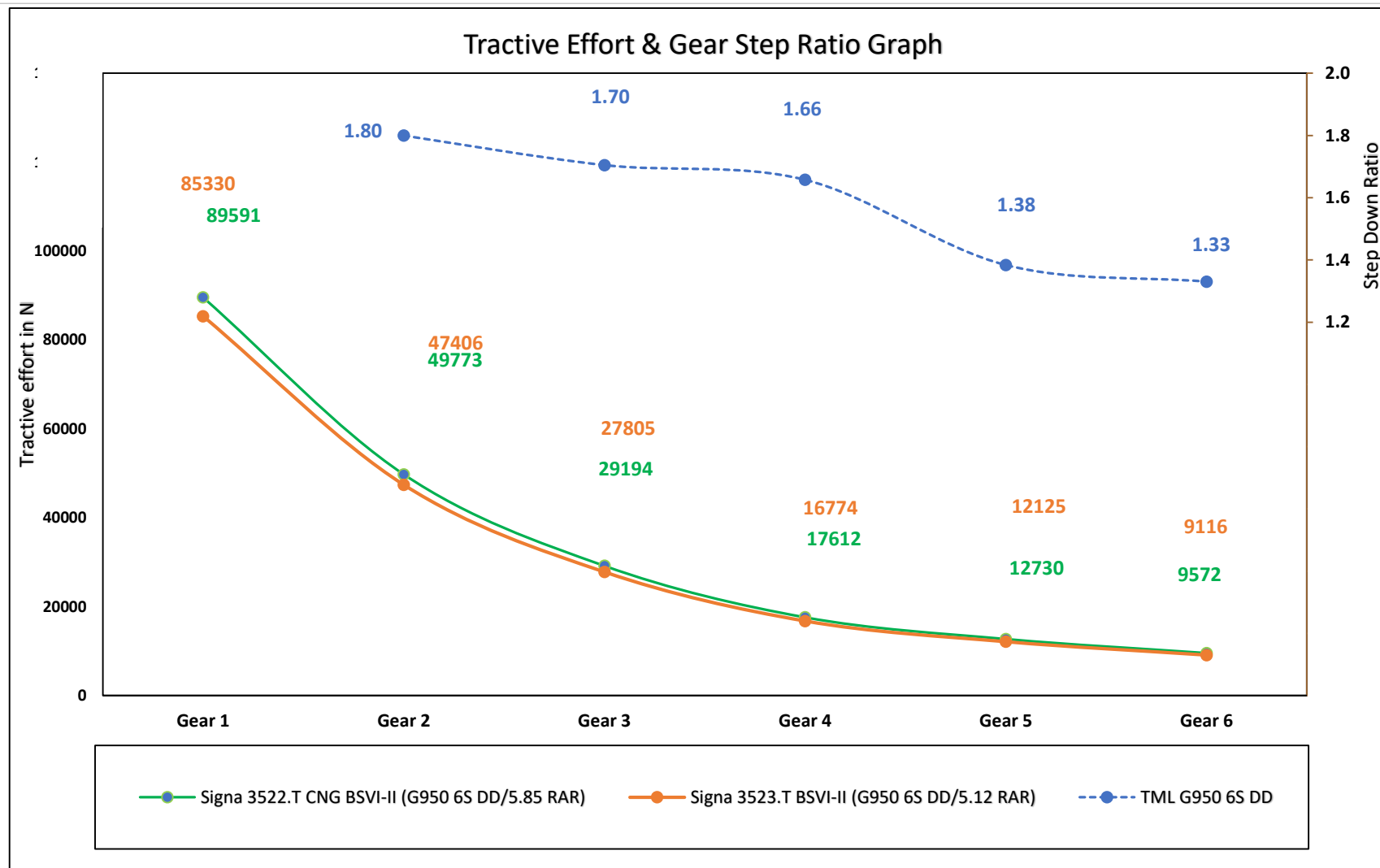
Executive FE summary of Signa 3522.T CNG BSVI-II vehicle, with AC ON & OFF, in comparison with Signa 3523.T BSVI-II Base vehicle with AC OFF.

PED | HCV | 14.07.2025



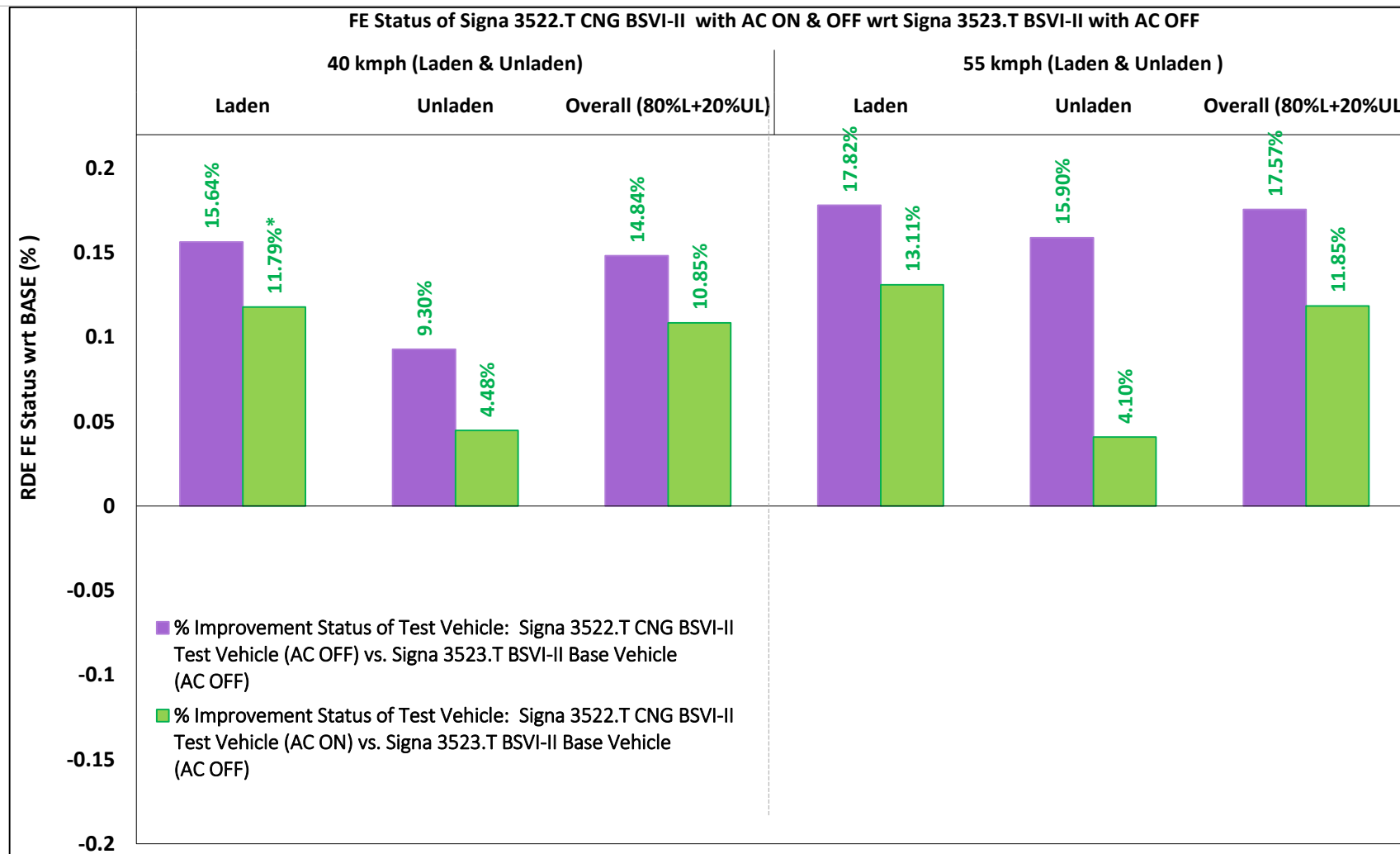
Sr. No.	Parameters	Base Vehicles BSVI-II	Test Vehicles BSVI-II
		Signa 3523.T BSVI-II	Signa 3522.T CNG BSVI-II
1	Engine	6 Cylinder 5.6 L / 230 HP BSVI	TATA 5.3 SGI Bharat Stage VI OBD-II
1.1	Engine Max. Power	220 HP @ 2300 rpm	220 HP @ 2300 rpm
1.2	Engine Max. Torque	925Nm @ 1000-1300rpm	850 Nm @ 1100 - 1600 rpm
1.3	Power to weight (HP/Ton)	6.28	6.28
1.4	Torque to weight (Nm/Ton)	26.42	24.28
1.5	Engine Calibration Number	XWTAA19D07	53SGITCBS6
1.6	Engine Calibration Number (Production)	PF10040	PF10040
1.7	Engine Calibration Maturity	100%	100%
1.8	Regeneration Strategy	Auto	NA
1.9	Urea Dosing	Yes	NA
2.1	EOL Parameters	LBSC - 1900,2000,35000 kg, VAM - 3.0,2.5, 10, 40 Kmph, GDP - 50, 50 Kmph, RSG – 80 Kmph	NA
2	Fuel Economy Improvement Technologies		
2.1	Engine Cooling Fan	E-Viscous	E-Viscous
2.2	Fuel Economy Switch	4TC	Yes
2.3	Gear Shift Adviser	Yes	Yes
2.4	Accelerator Pedal Map	Yes	Yes
3	Other Parameters		
3.1	Fuel Tank type	Polymer	Polymer
3.2	Driveline	G 950 DD , FGR 9.36	G 950 DD , FGR 9.36
3.3	Gear Ratio	I-9.36, II-5.2, III-3.05, IV-1.84, V-1.33, VI-1,	I-9.36, II-5.2, III-3.05, IV-1.84, V-1.33, VI-1,
3.4	RAR	5.12	5.85
3.5	Tire size	Front – Rib 295/90R 20 Bridgestone Tyre Rear – Lug 295/90R20 Bridgestone Tyre	Front – Rib 295/90R 20 Apollo Tyre Rear – Lug 295/90R20 Apollo Tyre

Tractive Effort Comparison :



Tractive effort comparison of Signa 3522.T CNG BSVI-II Vs Signa 3523.T BSVI-II vehicle

Fuel Economy Status



FE Status of Signa 3522.T CNG BSVI-II with AC ON & OFF wrt Signa 3523.T BSVI-II with AC OFF.

Fuel Economy summary at 40 kmph: AC ON & OFF

Sr. No.	Parameters		Base Vehicle												Test Vehicle															
			Signa 3523.T 5.6L BSVI-II												Signa 3522.T CNG BSVI-II															
1	Vehicle Identification No. (ERC No. /EJ No./Reg. No.)		ERC-1309												EL00313															
2	Vehicle GVW/GCW	kg	35000												35000															
	Unladen weight	kg	--												10500															
3	Engine Details (Make/cc/Norms/no. of Cyl.)		Cummins ISBe 5.6L/ Ashwamedh												TATA 5.3 SGI Bharat Stage VI OBD-II															
	Engine Max Power		220 HP @ 2300 rpm												220 HP @ 2300 rpm															
	Engine Max Torque		925 Nm @ 1000-1300 rpm												850 Nm @ 1100 - 1600 rpm															
	Engine Calibration No.		XWTAA19D07												53SGITCB56															
	Engine Calibration Maturity Level		As per production released												90%															
4	Regeneration Strategy (Time ase/Soot base)		Time Based (100 Hrs)												NA															
5	Gear Box (make/drive/FGR)		G950 6S DD, FGR- 9.36												G950 6S DD, FGR- 9.36															
	Gear Ratios		1-9.36, 2- 5.20, 3- 3.05, 4- 1.84, 5- 1.33, 6- 1.00												1-9.36, 2- 5.20, 3- 3.05, 4- 1.84, 5- 1.33, 6- 1.00															
6	RAR		5.12												5.85															
7	Tyre Details		Bridgestone 295/90R20 low CRR												Apollo Endu NRG 295/90R20															
8	LBSC	High & Low Engine Speed Breakpoint	RPM												NA															
		Vehicle Mass Threshold	kg												NA															
	VAM	Variable Acceleration 1, 2	kmph/sec												NA															
		Variable Acceleration Speed 1, 2	kmph												NA															
	GDP	GD Max. Veh. Speed (Heavy & Light Engine Load)	kmph												NA															
	RSG	Maximum Accelerator Vehicle Speed	kmph												--															
	Torque Factor	Torque Limit Medium Adjustment Factor	Balance												--															
		Torque Limit Maximum Adjustment Factor	Economy												--															
	PTP	Powertrain Protection (Enable / Disable)													NA															
		Torque Limit	Nm												NA															
9	USFE / Torque Table Revision (1/2/3/n)		USFE-1												--															
9	Fan Details (type/make/blades)		24" E-Viscous fan												24" E-Viscous fan															
FE Avenues (Engine Level)	Base Cal Improvement (Cal Maturity %)		100%												100% Cal Maturity															
	Low Viscous Engine Oil (TMGO-10W30)		Yes (15W40CK4+)												Yes (15W40)															
	FE Switch		IUSFE												Yes (3 Mode : E/B/P)															
	Engine Features (EOL : Enable/Disable)		Enable												No															
	ATS Optimization		Yes												No															
	APM		Yes												No															
	Gear based VAM		No												No															
	Intelligent mode shifter		No												No															
	Moving Mass Estimation		No												No															
	Driveline Optimization		No												NA															
FE Avenues (Vehicle Level)	GSA (Gear Shift Advisor)		Yes												Yes (Not Tuned)															
	MSA (Mode Shift Advisor)		NA												NA															
	Low CRR Tyres		Yes												Yes															
	Low Viscous rear axle oil (80W90)		Yes												Yes															
	Smart Alternator (LIN)		No												NA															
	THU/Unitised/Low friction bearing		No												NO															
	Route		Jamshedpur-Ranchi-Jamshedpur												Jamshedpur-Ranchi-Jamshedpur															
FE Trial Details	FE Duty Cycle (Laden/Unladen)		Laden Trial @ 40 kmph												Unladen Trial @ 40 kmph				Laden Trial @ 40 kmph AC OFF				40 kmph With AC ON (Max. Cooling)		40 kmph With AC ON (Mini. Cooling)		Unladen Trial @ 40 kmph AC OFF		Unladen Trial @ 40 kmph with AC ON (Min Cooling)	
	Trial Date		30.05.25	31.05.25	07.06.25	08.06.25	14.06.25	15.06.25	05.06.25	06.06.25	28.06.25	29.06.25	26.06.25	27.06.25	30.05.25	31.05.25	07.06.25	08.06.25	14.06.25	15.06.25	05.06.25	06.06.25	28.06.25	29.06.25	26.06.25	27.06.25				
	Trip Distance		Km	242.00	242.00	242.00	242.00	242.00	242.00	242.00	244.00	248.00	246.00	242.00	207.00	213.00	211.60	210.70	198.00	198.30	201.80	201.00	217.40	217.80	207.00	200.50				
	Total CNG quantity filled		Kg												50.00	50.00	50.00	50.00	50.00	50.00	50.00	50.00	30.00	30.00	30.00	30.00				
	Total Diesel Quantity		Litre	66.25	63.53	64.33	64.80	64.78	65.81	65.44	65.24	35.76	37.80	36.70	36.67															
	Total Urea Quantity		Litre	3.4	3.4	3.6	3.5	3.8	3.0	3.8	3.5	1.0	0.2	1.6	1.8															
	Equivalent Urea Quantity (50% Consumption)		Litre	1.70	1.70	1.80	1.75	1.90	1.50	1.90	1.75	0.50	0.10	0.80	0.90															
	Total Equivalent Fluid Consumed		Litre	67.95	65.23	66.13	66.55	66.68	67.31	67.34	66.99	36.26	37.90	37.50	37.57															
	Fuel Economy		Kmpl/Km	3.65	3.81	3.76	3.73	3.74	3.68	3.70	3.71	6.82	6.56	6.70	6.60															
	Equivalent Fluid Economy		Kmpl/Km	3.56	3.71	3.66	3.64	3.63	3.60	3.59	3.61	6.73	6.54	6.56	6.44	4.14	4.26	4.23	4.21	3.96	3.97	4.04	4.02	7.25	7.26	6.90	6.68			
	Status of Equivalent Fluid economy		%	Base	Base	Base	Base	Base	Base	Base	Base	Base	Base	Base	Base	16.25%	14.83%	15.65%	15.89%	9.11%	10.31%	12.31%	11.28%	7.69%	10.95%	5.18%	3.76%			
	Cluster Total Odo. Reading (at End of trial)		Km	254	254	254	254	254	254	253	254	256	260	258	252	215.5	221.0	219.6	218.70	205.0	205.3	208.8	208.0	225.4	225.8	216.0	208.5			
	Cluster Trip AFE		Kmpl/Km	3.90	4.00	4.00	4.00	3.90	4.00	3.90	3.90	7.00	6.70	7.00	6.90	4.4	4.5	4.5	4.5	4.1	4.2	4.3	4.0	7.9	7.9	6.4	6.2			
	Avg. Equivalent Fluid Economy			3.64				3.61		3.60		6.64		6.50		4.21				3.96		4.03		7.25		6.79				
	Status of Fluid Economy			Base												15.64%				9.71%		11.79%		9.30%		4.48%				
Status of AC ON wrt AC OFF			Base																		-4.36%		-8.38%							
Overall Avg. Fuel Economy (kmpl/kmpkg) : L + UL			4.00						3.96		4.60										4.38									
Overall status of Avg. Fuel Economy (kmpl/kmpkg) : L + UL			80% + 20%				80% + 20%		Base				14.84%				10.85%													

Fuel Economy summary at 55 kmph: AC ON & OFF

Sr. No.	Parameters		Base Vehicle												Test Vehicle												
			Signa 3523.T 5.6L BSVI-II												Signa 3522.T CNG BSVI-II												
1	Vehicle Identification No. (ERC No./EI No./Reg. No.)		ERC-1309												EL00313												
2	Vehicle GVW/GCW	kg	35000												35000												
	Unladen weight	kg	--												10500												
3	Engine Details (Make/cc/Norms/no. of Cyl.)		Cummins ISBe 5.6L/ Ashwamedh												TATA 5.3 SGI Bharat Stage VI OBD-II												
	Engine Max Power		220 HP @ 2300 rpm												220 HP @ 2300 rpm												
	Engine Max Torque		925 Nm @ 1000-1300 rpm												850 Nm @ 1100 - 1600 rpm												
	Engine Calibration No.		XWTAA19D07												53SGITCB56												
	Engine Calibration Maturity Level		As per production released												90%												
4	Regeneration Strategy (Time ase/Soot base)		Time Based (100 Hrs)												NA												
5	Gear Box (make/drive/FGR)		G950 6S DD, FGR- 9.36												G950 6S DD, FGR- 9.36												
6	Gear Ratios		1-9.36, 2- 5.20, 3- 3.05, 4- 1.84, 5- 1.33, 6- 1.00												1-9.36, 2- 5.20, 3- 3.05, 4- 1.84, 5- 1.33, 6- 1.00												
7	RAR		5.12												5.85												
8	Tyre Details		Bridgestone 295/90R20 low CRR												Apollo Endu NRG 295/90R20												
8	LBSC	High & Low Engine Speed Breakpoint	RPM		1900 rpm,2000 rpm										NA												
		Vehicle Mass Threshold	kg		35000										NA												
	VAM	Variable Acceleration 1, 2	kmph/sec		3.0,2.5										NA												
		Variable Acceleration Speed 1, 2	kmph		10, 40										NA												
	GDP	GD Max. Veh. Speed (Heavy & Light Engine Load)	kmph		50, 50										NA												
	RSG	Maximum Accelerator Vehicle Speed	kmph		80										--												
	Torque Factor	Torque Limit Medium Adjustment Factor	Balance		130%										--												
		Torque Limit Maximum Adjustment Factor	Economy		120%										--												
	PTP	Powertrain Protection (Enable / Disable)			Disable										NA												
		Torque Limit	Nm		NA										NA												
9	USFE / Torque Table Revision (1/2/3/n)		USFE-1												--												
9	Fan Details (type/make/blades)		24" E-Viscous fan												24" E-Viscous fan												
FE Avenues (Engine Level)	Base Cal Improvement (Cal Maturity %)		100%												100% Cal Maturity												
	Low Viscous Engine Oil (TMGO-10W30)		Yes (15W40CK4+)												Yes (15W40)												
	FE Switch		IUSFE												Yes (3 Mode : E/B/P)												
	Engine Features (EOL : Enable/Disable)		Enable												No												
	ATS Optimization		Yes												No												
	APM		Yes												No												
	Gear based VAM		No												No												
	Intelligent mode shifter		No												No												
	Moving Mass Estimation		No												No												
	FE Avenues (Vehicle Level)	Driveline Optimization		No												NA											
GSA (Gear Shift Advisor)		Yes												Yes (Not Tuned)													
MSA (Mode Shift Advisor)		NA												NA													
Low CRR Tyres		Yes												Yes													
Low Viscous rear axle oil (80W90)		Yes												Yes													
Smart Alternator (LIN)		No												NA													
THU/Unitised/Low friction bearing		No												NO													
Route		Jamshedpur-Ranchi-Jamshedpur												Jamshedpur-Ranchi-Jamshedpur													
FE Trial Details	FE Duty Cycle (Laden/Unladen)		Laden Trial @ 55 Kmph								Unladen Trial @ 55 Kmph				Laden Trial @ 55 Kmph AC OFF				5S Kmph With AC ON (Max. Cooling)		5S Kmph With AC ON (Mini. Cooling)		Unladen Trial @ 55 Kmph AC OFF		Unladen Trial @ 55 Kmph with AC ON (Min Cooling)		
	Trial Date		28.05.25	29.05.25	12.06.25	13.06.25	01.06.25	02.06.25	10.06.25	11.06.25	30.06.25	01.07.25	23.06.25	25.06.25	28.05.25	29.05.25	12.06.25	13.06.25	10.06.25	11.06.25	30.06.25	01.07.25	23.06.25	25.06.25			
	Trip Distance		Km	242.00	242.00	242.00	242.00	242.00	242.00	242.00	248.00	248.00	248.00	248.00	200.70	196.50	201.00	201.00	182.40	183.60	195.20	195.10	199.00	200.00	240.00	211.00	
	Total CNG quantity filled		Kg												49.94	50.00	50.00	50.00	48.72	50.00	50.00	50.00	30.00	30.00	40.00	35.00	
	Total Diesel Quantity		Litre	70.46	70.00	67.78	69.84	66.97	69.76	67.70	68.88	42.40	42.00	42.17	42.43												
	Total Urea Quantity		Litre	3.4	3.6	3.8	3.8	3.5	3.8	3.6	3.8	2.0	2.1	1.2	1.3												
	Equivalent Urea Quantity (50% Consumption)		Litre	1.70	1.80	1.90	1.90	1.75	1.90	1.80	1.90	1.00	1.05	0.60	0.65												
	Total Equivalent Fluid Consumed		Litre	72.16	71.80	69.68	71.74	68.72	71.66	69.50	70.78	43.40	43.05	42.77	43.08												
	Fuel Economy		Kmpl	3.43	3.46	3.57	3.47	3.61	3.47	3.57	3.51	5.85	5.90	5.88	5.84												
	Equivalent Fluid Economy		Kmpl/Kmpkg	3.35	3.37	3.47	3.37	3.52	3.38	3.48	3.42	5.71	5.76	5.80	5.76	4.02	3.93	4.02	4.02	3.74	3.67	3.90	6.63	6.67	6.00	6.03	
	Status of Equivalent Fluid economy		%	Base	Base	Base	Base	Base	Base	Base	Base	Base	Base	Base	Base	19.83%	16.60%	15.75%	19.17%	6.31%	8.73%	12.12%	14.13%	16.08%	15.73%	3.48%	4.72%
	Cluster Total Odo. Reading (at End of trial)		Km	254	254	254	254	254	254	254	254	260	260	260	260	207.7	203.5	208.0	208.0	189.4	190.7	202.2	202.1	207.5	208.0	248.0	218.0
	Cluster Trip AFE		Kmpl/Kmpkg	3.70	3.80	3.80	3.70	3.80	3.70	3.80	3.70	6.00	6.00	6.20	6.20	4.1	4.1	4.2	4.2	3.8	3.7	4.1	4.2	7.0	7.2	6.4	6.4
Avg. Equivalent Fluid Economy			3.39				3.45		3.45		5.74		5.78		4.00				3.71		3.90		6.65		6.01		
Status of Fluid Economy			Base												17.82%				7.50%		13.11%		15.90%		4.10%		
Status of AC ON wrt AC OFF			Base																		-2.36%				-9.66%		
Overall Avg. Fuel Economy (kmpl/kmpkg) : L + UL			80% + 20%				3.69				3.75				4.34								4.20				
Overall status of Avg. Fuel Economy (kmpl/kmpkg) : L + UL			80% + 20%				Base								17.57%								11.85%				

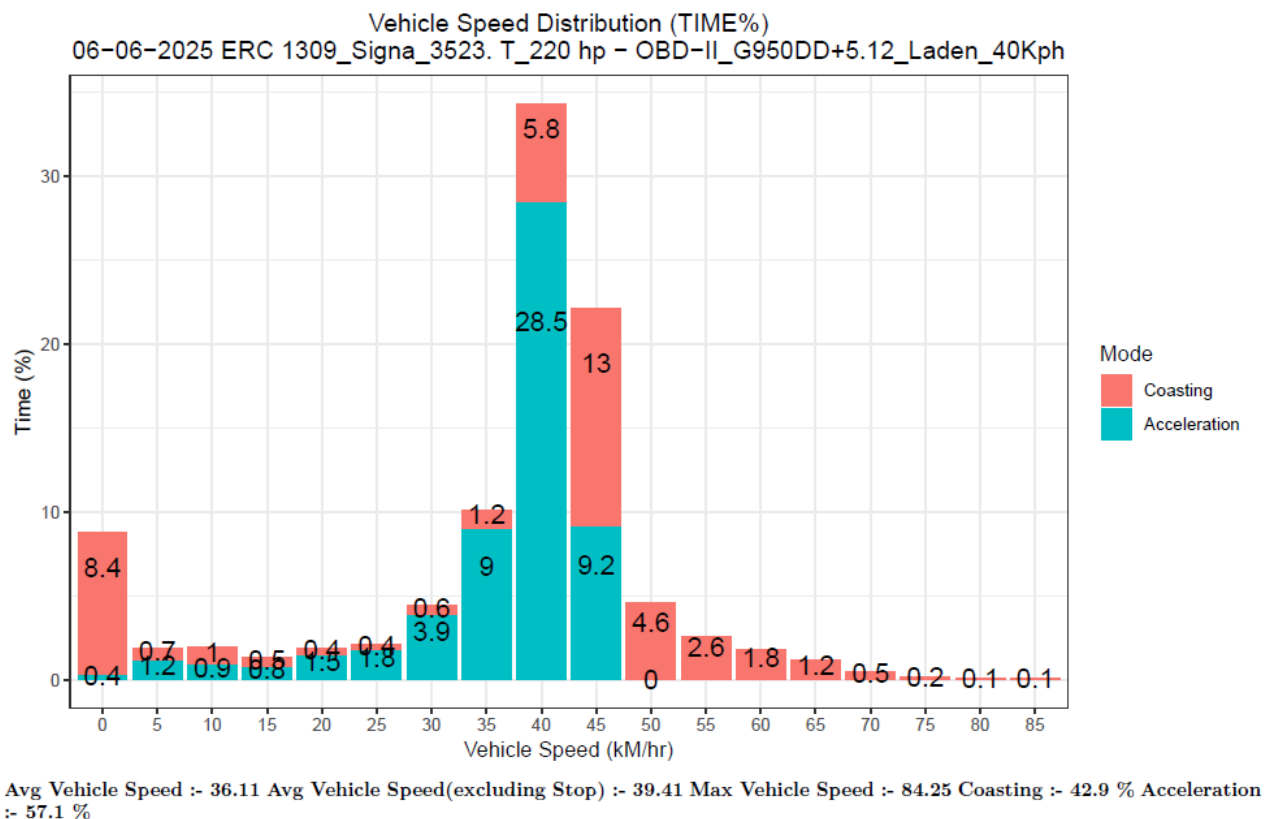
Duty Cycle Analysis Jamshedpur-Ranchi Road FE Trial

Parameters Comparison

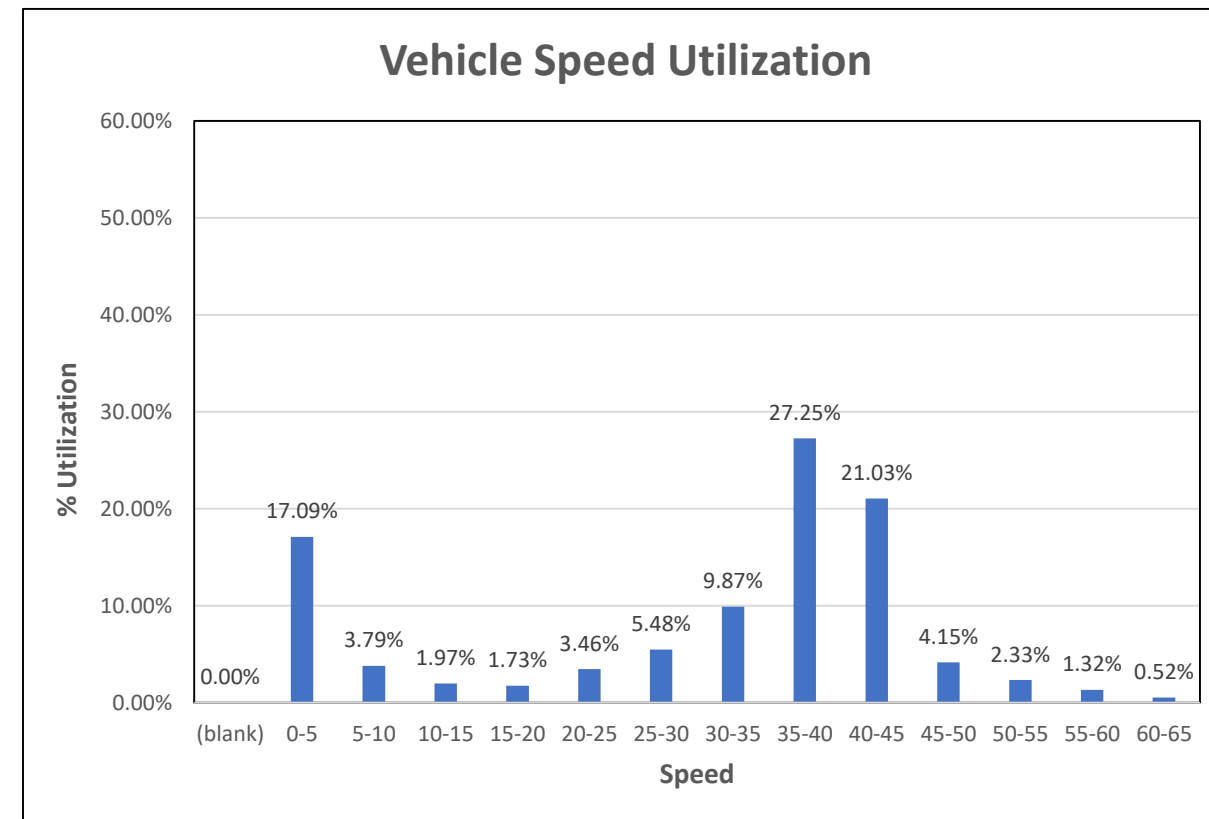
Sr. No.	Parameter	Unit	Duty Cycle Analysis	
			Laden : Highway Route (05/06/2025)	
			Signa 2820.T	Signa 3522.T
1	Average Engine Speed	RPM	1093.09	1132.65
	Avg. Engine Speed @ Top gear	RPM	--	1213.6
2	Average Power (Positive Power)	Kw	36.34	--
	Average Power@ Top gear	Kw	--	--
3	Average Torque	Nm	--	237.54
	Average Torque @ Top gear	Nm	--	238.23
4	Average Vehicle Speed	kmph	36.11	30.01
	Average Vehicle Speed @ Top gear	kmph	39.41	39.84
	Coasting	%	42.9	48.37
5	Idling Time (Vehicle speed)	%	8.6	12.26
6	Full Throttle	%	23.3	2.09
7	Time in Top gear	%	65.08	65.12
8	Time in gear down	%	14.61	6.94
9	Gear shifts per 100 km	Nos	114.00	--
10	Eco Mode Utilization	%	75.85	56.91
11	Compressor Engagement	%	--	40.19
12	Distance	Km	228	208

Vehicle Speed Utilization

Signa 3523.T BSVI-II



Signa 3522.T BSVI-II CNG

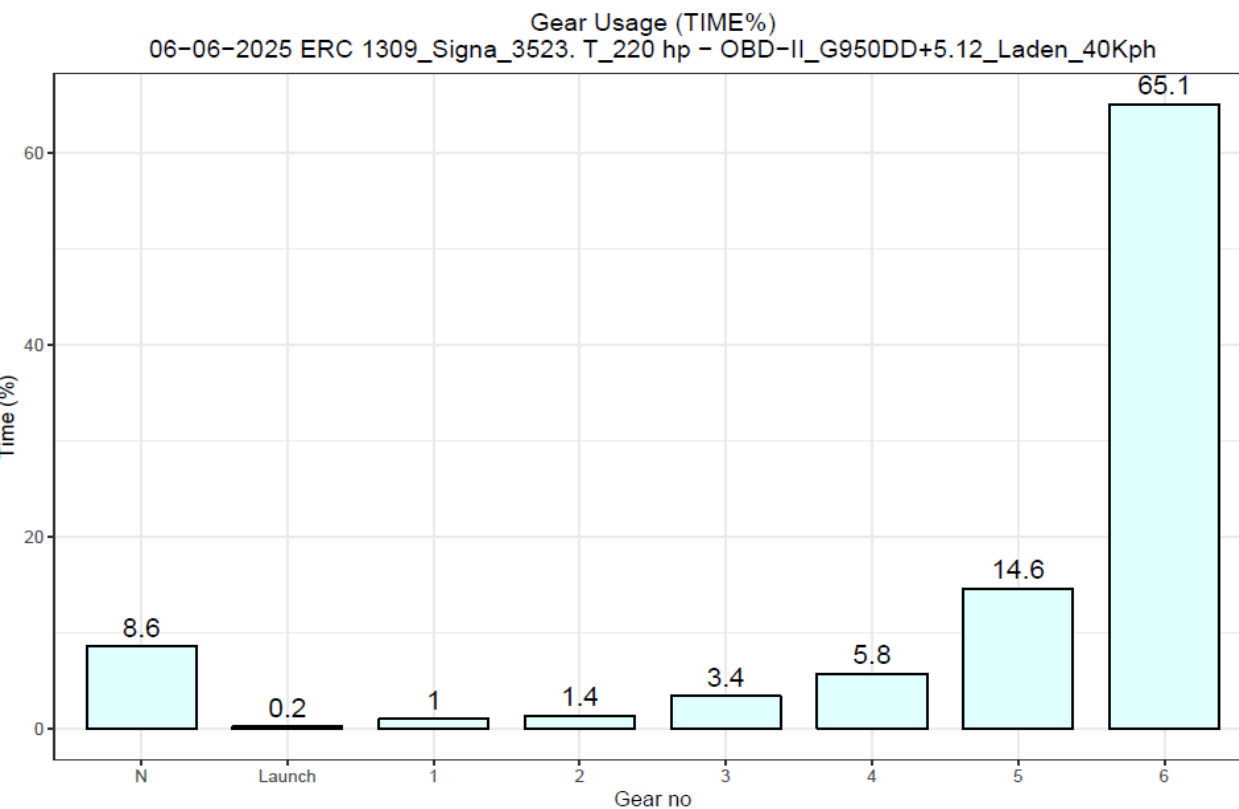


Remark:

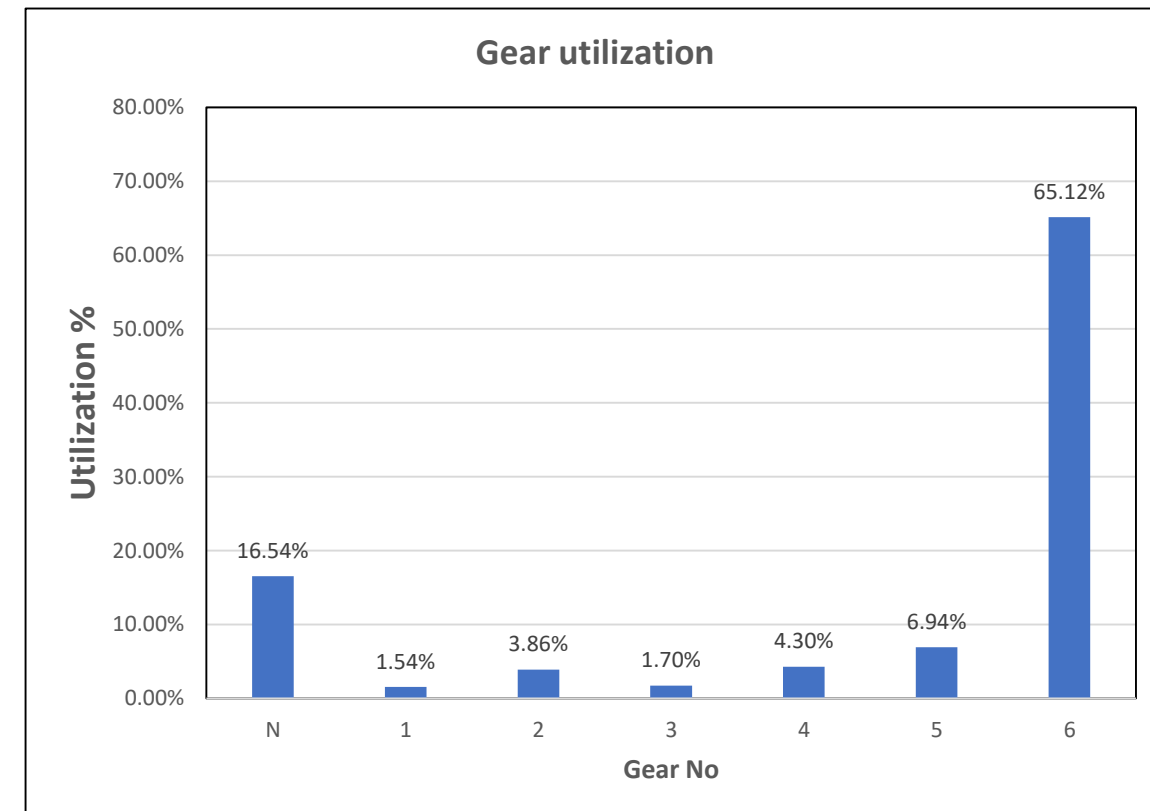
% Utilization in (35-45) speed range is **57.59% & 48.28%** in Signa 2820.T CNG & Signa 3522.T CNG respectively.

Gear Utilization

Signal 3523.T BSVI-II

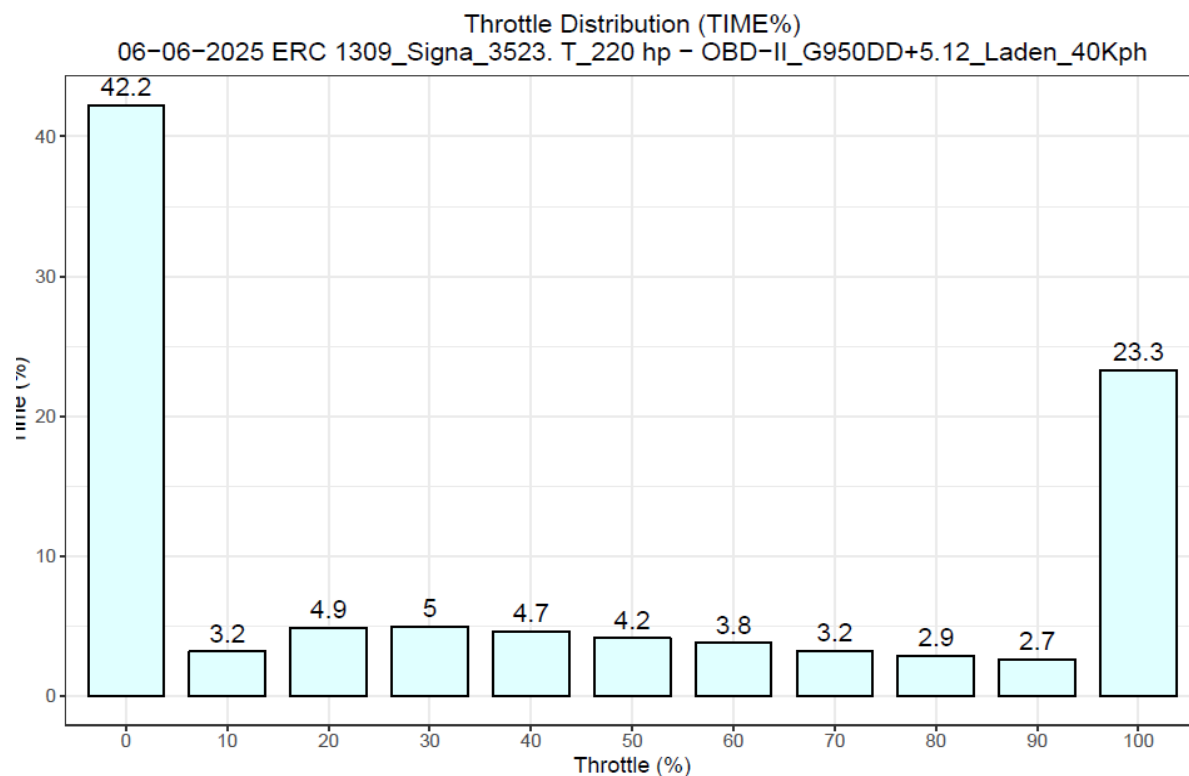


Signal 3522.T BSVI-II CNG

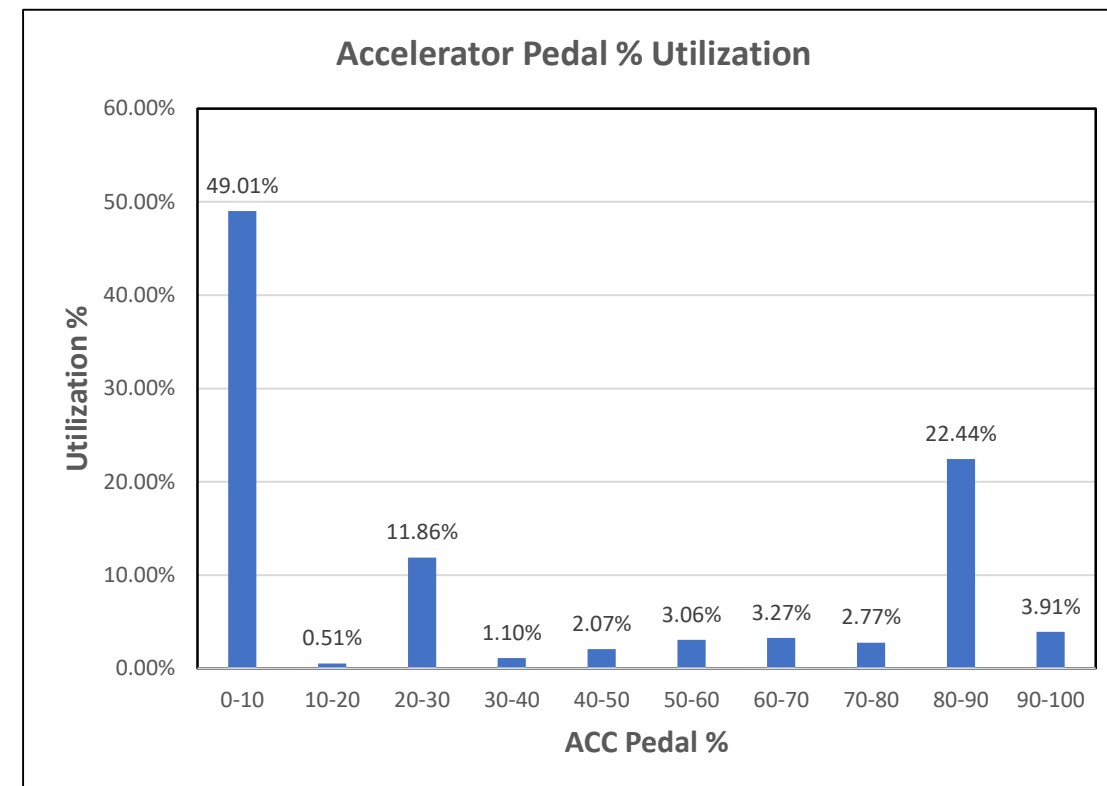


Accelerator Pedal % Utilization

Signa 3523.T BSVI-II

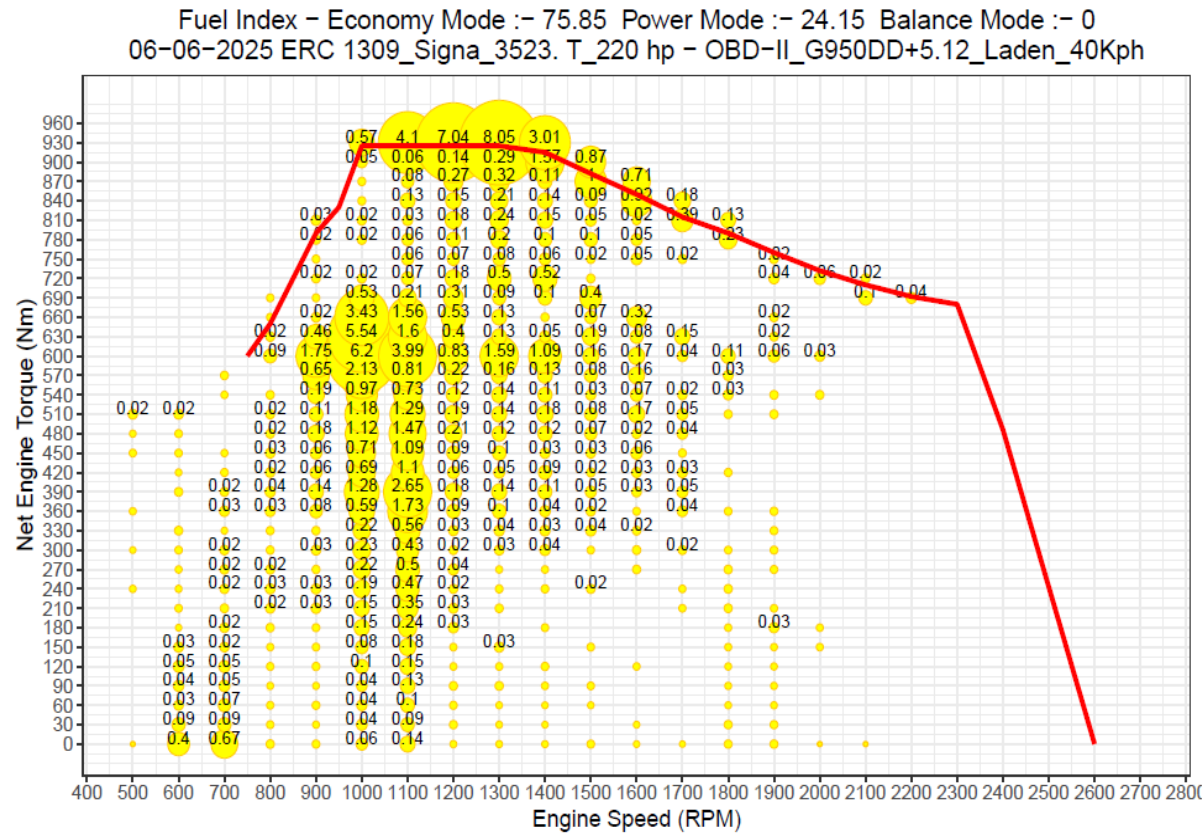


Signa 3522.T BSVI-II CNG

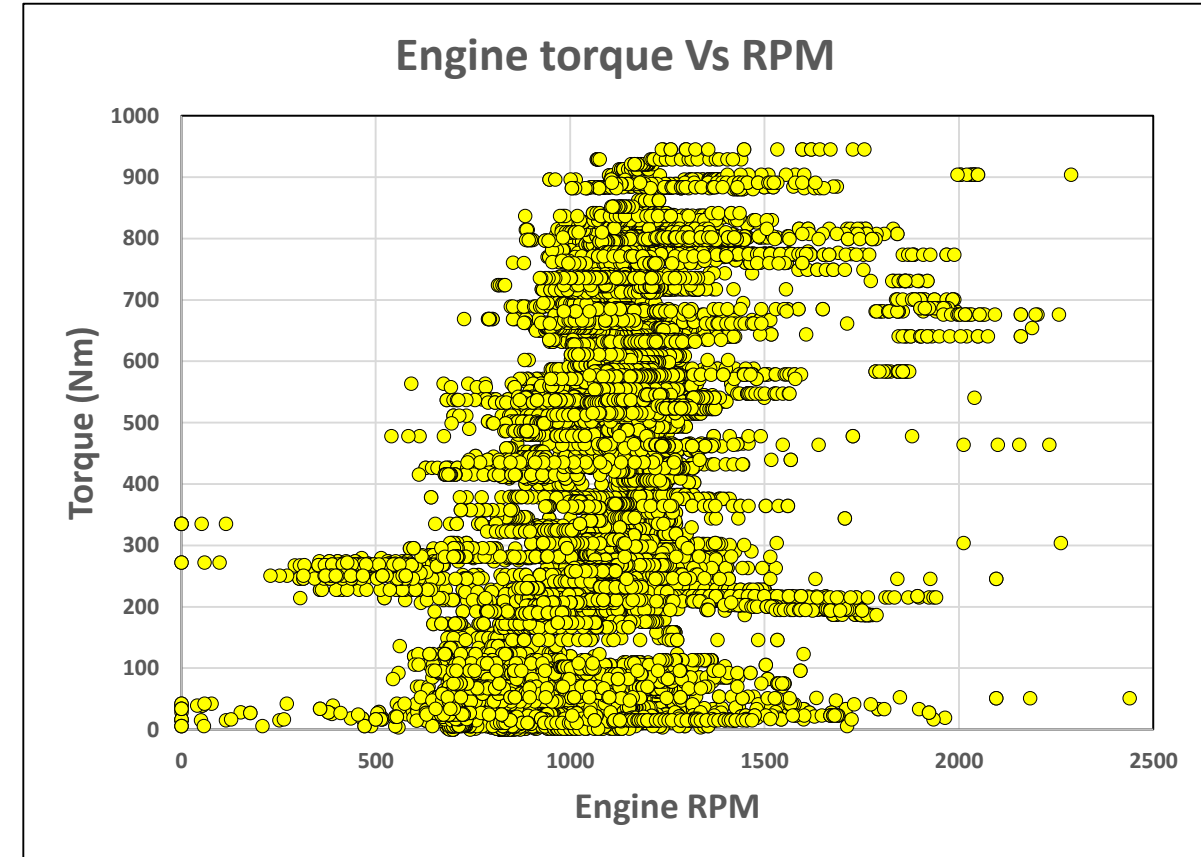


Torque Vs Engine RPM

Signal 3523.T BSVI-II

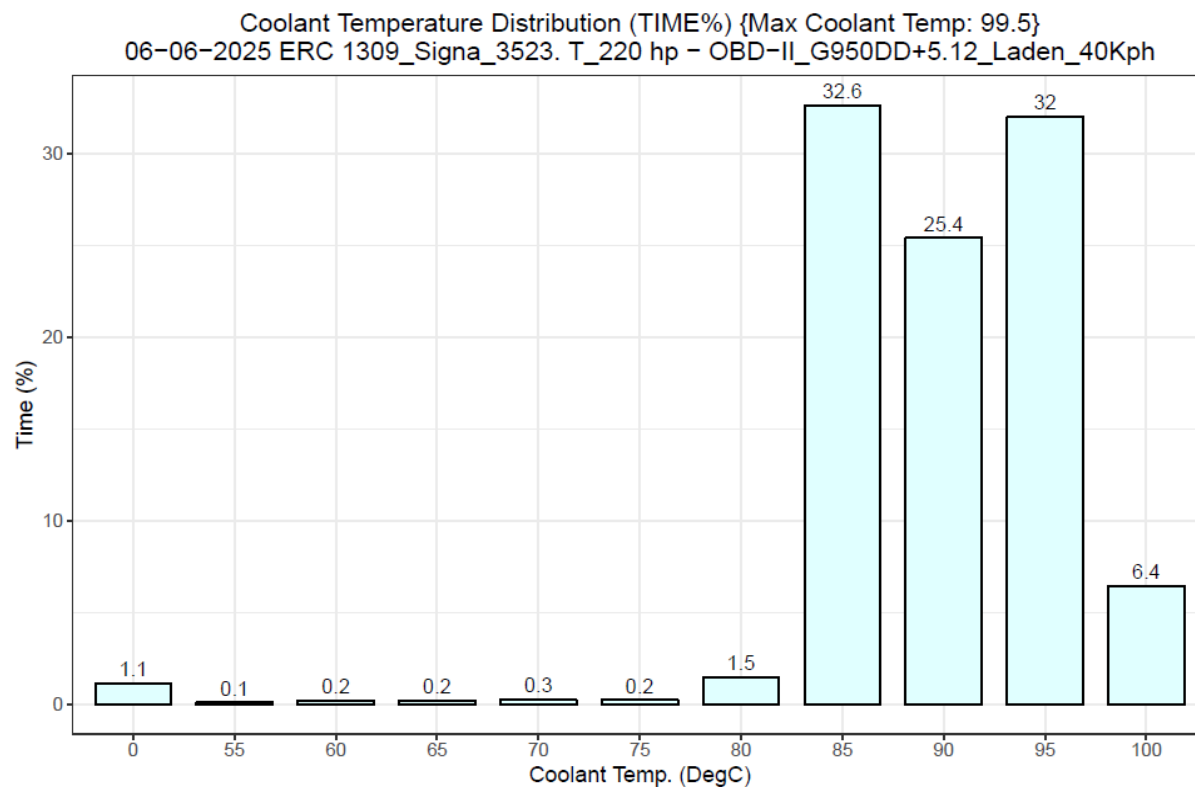


Signal 3522.T BSVI-II CNG

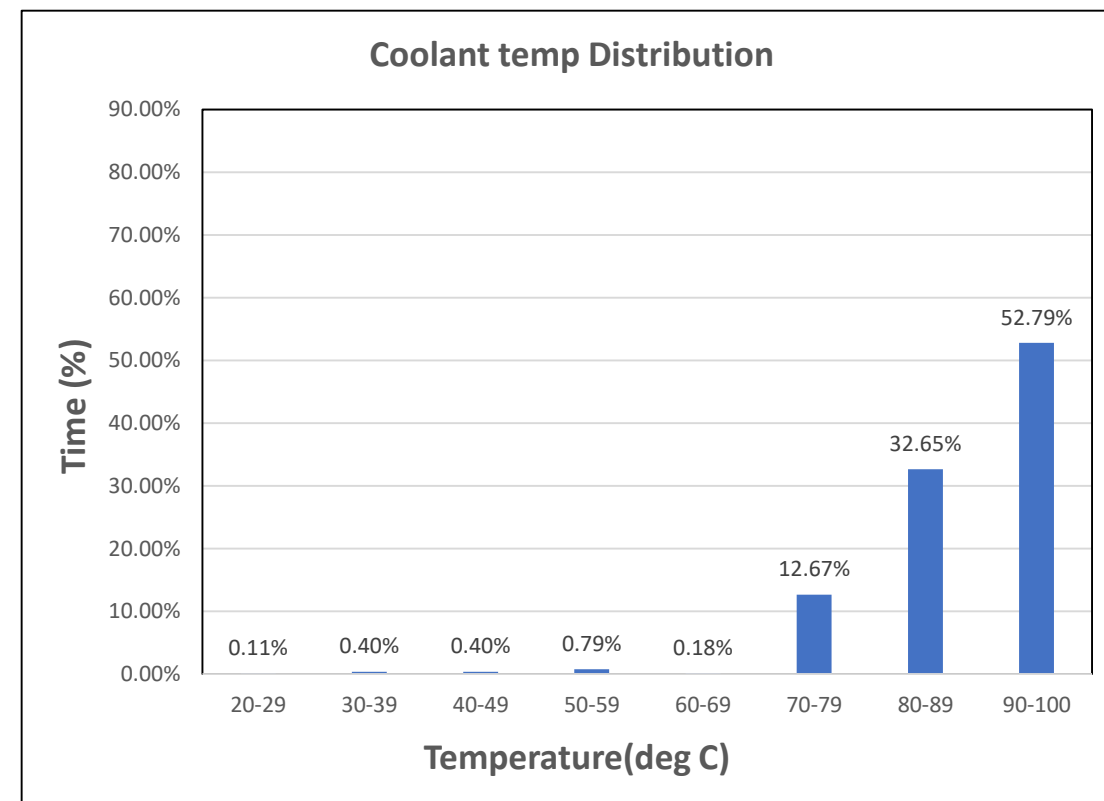


Coolant temp Distribution

Signa 3523.T BSVI-II

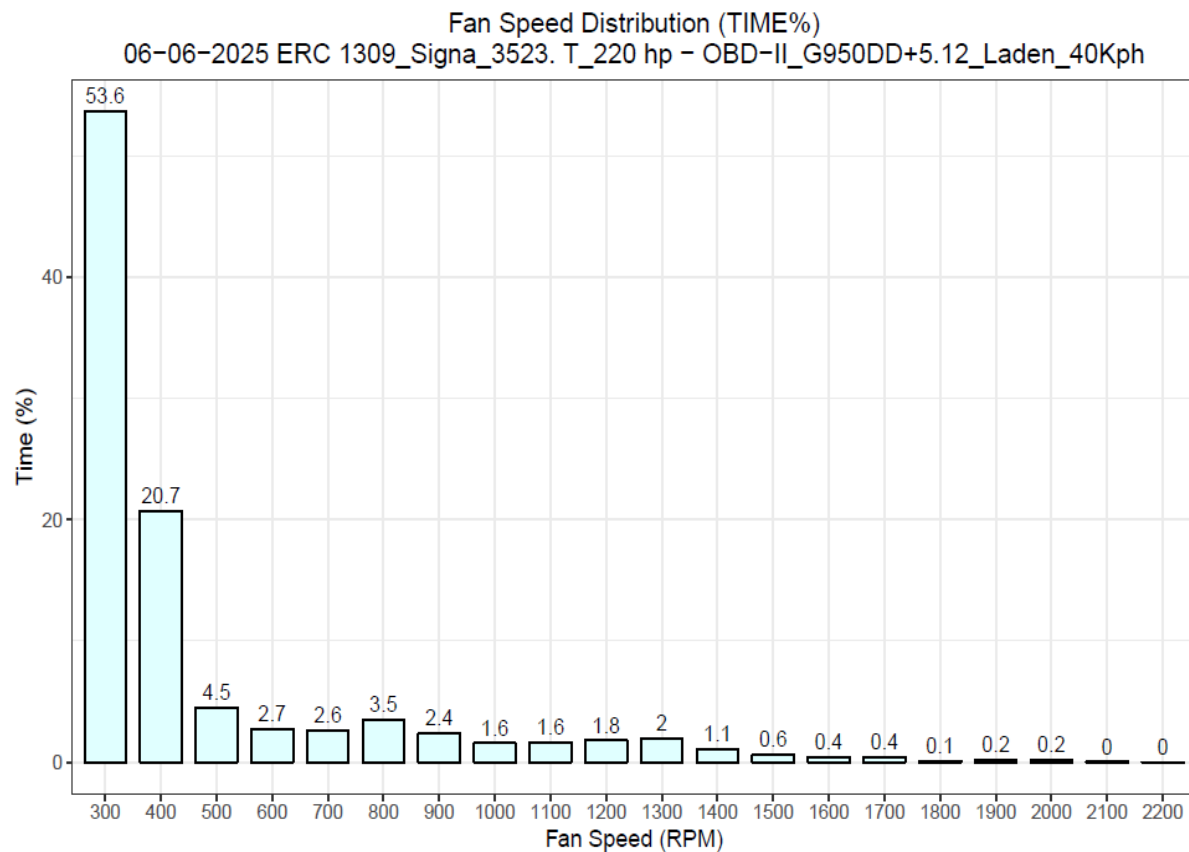


Signa 3522.T BSVI-II CNG

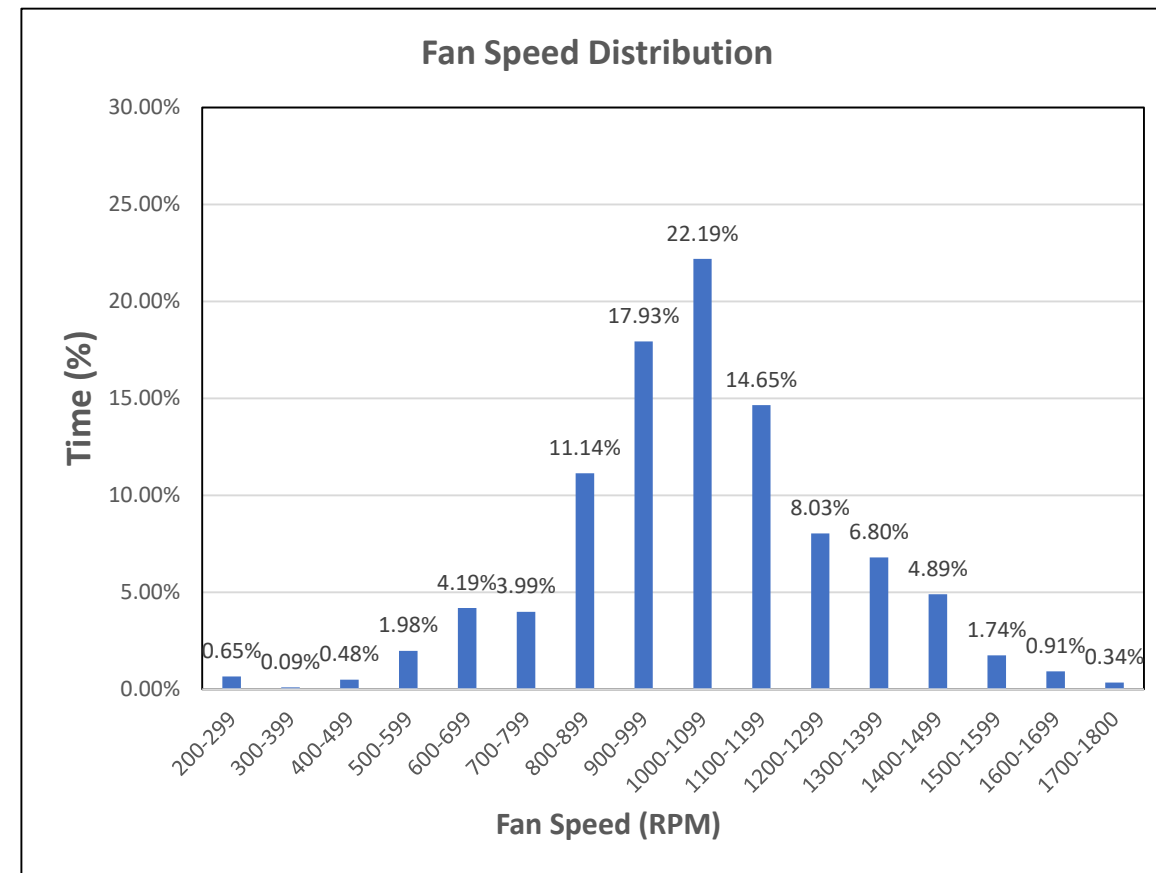


Fan Speed Distribution

Signa 3523.T BSVI-II



Signa 3522.T BSVI-II CNG



Fuel Economy Comparison summary

Vehicle Model for X-axis	Crusing speed	Duty Cycle	FLE Status of Signa 3523.T BSVI-II Base Vehicle with AC OFF	FLE Status of Signa 3522.T CNG BSVI-II Test Vehicle with AC OFF	FLE Status of Signa 3523.T BSVI-II Base Vehicle with AC OFF	FLE Status of Signa 3522.T CNG BSVI-II Test Vehicle with AC ON	% Improvement Status of Test Vehicle: Signa 3522.T CNG BSVI-II Test Vehicle (AC OFF) vs. Signa 3523.T BSVI-II Base Vehicle (AC OFF)	% Improvement Status of Test Vehicle: Signa 3522.T CNG BSVI-II Test Vehicle (AC ON) vs. Signa 3523.T BSVI-II Base Vehicle (AC OFF)	Status of % Delta of Signa 3522.T CNG BSVI-II Vehicle AC ON wrt AC OFF
FE Status of Signa 3522.T CNG BSVI-II with AC ON & OFF wrt Signa 3523.T BSVI-II with AC OFF	40 kmph (Laden & Unladen)	Laden	3.64	4.21	3.60	4.03	15.64%	11.79%	-4.4%
		Unladen	6.64	7.25	6.50	6.79	9.30%	4.48%	-6.4%
		Overall (80%L+20%UL)	4.00	4.60	3.96	4.38	14.84%	10.85%	-4.6%
	55 kmph (Laden & Unladen)	Laden	3.39	4.00	3.45	3.90	17.82%	13.11%	-2.4%
		Unladen	5.74	6.65	5.78	6.01	15.90%	4.10%	-9.6%
		Overall (80%L+20%UL)	3.69	4.34	3.75	4.20	17.57%	11.85%	-3.4%

Thank You

TEST REQUISITION

Ref.	-	Date:	11/07/2025
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To,

ERC Testing

Kindly arrange to take up for testing of the following component.

1	Part Description	TATA SIGNA 3522.T BSVI CNG			
2	Part Number	52010160R,			
3	Drawing number	52010160R	Mod status	G;2,	Mod Date 06/06/2025
4	Vehicle Model / Project	Request from Mr.Amit Gupta, Mr.Amit Kumar Dhayade, Dated 14.09.2022			
5	WBS Number	P.23.EA.5045.3L CNG RDE Engine on M&HCV Truck Segme			
6	Supplier	TML			
7	Number of samples	-			
8	Inspection report	Enclosed	✓	If No commitment date:	
9	Material report	Enclosed	✓	If No commitment date:	
10	Suppliers test report	Enclosed	✓	If No commitment date:	
11	Reason for testing	First Source Development	✗	Weight Reduction	✗
		Alternate source development	✗	Cost Reduction	✗
		Data generation Test	✗	Regulatory test	✗
		Any other (pl. specify)	✗	New Development	✓
12	What testing required?				
13	Remarks	Test request for TATA Signa 3522.T BSVI CNG FE			
14	Peripheral Parts Availability	No			

For Testing Dept Use only	Test Item Number and Date(stamp)	Test Engineer

Authorized Signature:	
Name:	
Dept:	

Contact Person:	Patil Shubham
Phone Number	

TATA MOTORS LIMITED ERC PUNE	TEST PLAN		DATE: 11/08/2025					
	VEHICLE TESTING DEPARTMENT							
Description: Fuel Economy evaluation of Signa 3522.T CNG BSVI-II					Test Item No: UA001757			
					New	✓	Moderate	✗
					Major	✗	Minor	✗

Part No: 52010160R,		Modification No: G;2,	Enclosures: Annexure 1-5
Drawing No: 52010160R			
Make: TML		WBS Number: P.23.EA.504- 5.3L CNG RDE Engine on M&HCV Truck Segme	Copies to :
Vehicle Type: Request from Mr.Amit Gupta, Mr.Amit Kumar Dhayade, Dated 14.09.2022			
Test Requested by:			Equipments:
Test Requisition Reference:			
Reason for Test/s:	New development		
Regulatory Requirement:	a) Component: NA		
	b) Vehicle: TST/VT/WG/158		

Sr. No.	Test description	Test specification	Acceptance criteria	No. of Samples	Test Responsibility	Proto Phase (Mule/ Alpha/ Beta/PP)	Remarks
1	FE evaluation	TST/VT/WG/158	AS Per PED standard	1	Palash Das	Beta	completed

REMARKS: ESO		INTERNAL REF:	
TEST TO FAILURE: No			
MEOST: No			
RELIABILITY TESTING: No			
BENCHMARKING TESTING: No			
DFMEA/FMEA DONE: No			
PREPARED BY		CHECKED BY	APPROVED BY
Palash Das			